Feynman was driven to develop a deep theoretical understanding of nature, but he always remained close to the real and often grubby world of experimental results. Nobody who watched the elderly Feynman elucidate the cause of the Challenger space shuttle disaster by dipping an elastic band in ice water could doubt that here was both a showman and a very practical thinker.

Initially, Feynman made a name for himself from his work on the theory of subatomic particles, specifically the topic known as quantum electrodynamics or QED. In fact, the quantum theory began with this topic. In 1900, the German physicist Max Planck proposed that light and other electromagnetic radiation, which had hitherto been regarded as waves, paradoxically behaved like tiny packets of energy, or "quanta," when interacting with matter. These particular quanta became known as photons. By the early 1930s the architects of the new quantum mechanics had worked out a mathematical scheme to describe the emission and absorption of photons by electrically charged particles such as electrons. Although this early formulation of QED enjoyed some limited success, the theory was clearly flawed. In many cases calculations gave inconsistent and even infinite answers to well-posed physical questions. It was to the problem of constructing a consistent theory of QED that the young Feynman turned his attention in the late 1940s.

To place QED on a sound basis it was necessary to make the theory consistent not only with the principles of quantum mechanics but with those of the special theory of relativity too. These two theories come with their own distinctive mathematical machinery, complicated systems of equations that can indeed be combined and reconciled to yield a satisfactory description of QED. Doing this was a tough undertaking, requiring a high degree of mathematical skill, and was the approach followed by Feynman's contemporaries. Feynman himself, however, took a radically different route—so radical, in fact, that he was more or less able to write down the answers straightaway without using any mathematics!

To aid this extraordinary feat of intuition, Feynman invented a simple system of eponymous diagrams. Feynman diagrams are a symbolic but powerfully heuristic way of picturing what is going on when electrons, photons, and other particles interact with each other. These days Feynman diagrams are a routine aid to calculation, but in the early 1950s they marked a startling departure from the traditional way of doing theoretical physics.

The particular problem of constructing a consistent theory of quantum electrodynamics, although it was a milestone in the development of physics, was just the start. It was to define a distinctive Feynman style, a style destined to produce a string of important results from a broad range of topics in physical science. The Feynman style can best be described as a mixture of